
EM 656: Materials for Nuclear Power Plants

Course Code	EM 656
Course Name	Materials for Nuclear Power Plants
L-T-P-C	3-0-0-3
Intended for	M.Tech Energy Engineering
Distribution	Specialisation elective
Approval	9th Senate (May 2016), in the M.Tech Energy Engineering (Mechanical Design) curriculum

Course Modules

- **Module 1:** Fundamentals – binding energy, types of radiation and radioactive decay, neutron classification; Nuclear reactions – interaction of neutrons with matter, scattering, neutron capture and activation, Fission, neutron cross section and fusion, Nuclear reactors – Generation-I reactor (Magnox), Generation II reactors (PWR, BWR, AGR etc.), Generation III and IV reactors, Radiation damage – knock on atoms and displacement cascade, dislocation loops, nucleation of cavities and voids. (11)
- **Module 2:** Nuclear fuel – Metallic fuels (Uranium, Plutonium and thorium)-structure, thermal expansion, thermal conductivity, mechanical properties, irradiation growth, irradiation swelling and irradiation creep; Ceramic fuels (UO₂, Carbide and nitride), Cladding – Austenitic stainless steel – helium production, void swelling and its minimisation, creep and mechanical properties; Ferritic alloys; Zirconium alloys – irradiation growth, creep, mechanical properties and corrosion. (14)
- **Module 3:** Moderators – Graphite - structure, effect of irradiation on properties and Wigner energy; ZrH, Beryllium; liquid moderators, Control rods – materials used for control rods, Corrosion of structural components – stress corrosion cracking, (SCC), irradiation assisted SCC – radiolysis of water, persistent effects of stress, segregation and hardening. (17)